

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED



(15x2=30)

- Q.1. Answer the following short questions.
- Define population and sample.
 - Define tabulation. What are the necessary parts of a table?
 - For which type of data, the Geometric mean and harmonic mean are suitable?
 - The mean of 10 numbers is 8. If an eleventh number is included, the mean becomes 9. What is the value of the eleventh number?
 - Define the term dispersion. Write any three absolute measures of dispersion.
 - Differentiate between symmetry and skewness.
 - If sum of squared deviation from mean for 80 values is 1600, what will be the variance standard deviation and coefficient of variation if $\bar{x} = 20$?
 - Differentiate between simple and composite index numbers.
 - Differentiate between un-weighted index and weighted index number.
 - Differentiate between the term regression and correlation.
 - What is the relationship between regression coefficients and correlation coefficients? Find the correlation coefficient, if the regression coefficients are 1.2 and 0.60.
 - What is meant by coefficient of determination?
 - Define the term rank correlation, also write down the formula of rank correlation.
 - Define the term time series. Write down the components of a time series.
 - Differentiate between cyclical variations and sessional variations.

Solve the following questions.

- Q.2 (a) The following data relate to size of shoes sold at a store during a given week. Find the median and the modal size of shoes also find Q_1 and P_{90} . (6+4=10)

Size of Shoes	5	$5\frac{1}{2}$	6	$6\frac{1}{2}$	7	$7\frac{1}{2}$	8	$8\frac{1}{2}$	9	$9\frac{1}{2}$
No of Pairs	3	7	20	30	60	50	25	17	8	2

- (b) An analysis of the monthly wages paid to workers in two firms A and B belonging to the same industry gives the following results:

	Firm A	Firm B
No. of wage earners	120	145
Average monthly wages (Rs.)	28620	29540
Variance of wages	1250	1400

- Which firm A or B pays the larger amount of monthly wages?
- In which firm A or B is there greater relative variability in individual wages?

Page 1 of 2

P.T.O

CS CamScanner

- Q.3 (a) Calculate the coefficient of correlation from the following data.

X	3	4	5	6	7	8	9	10	11	12
Y	25	24	20	20	19	17	16	13	10	6

(4+6=10)

- (b) Construct index numbers from the following data by applying (i) Laspeyres's method (ii) Paasche's method.

Commodities	Base Year 1950		Current Year 1955	
	Prices	Quantities	Prices	Quantities
A	6	50	10	56
B	2	100	2	120
C	4	60	6	60
D	10	30	12	24
E	8	40	12	36

- Q.4 (a) Compute Seasonal Indices by link-relative method from the following data



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (15x2=30)

- Differentiate between Parameter and Statistic
- Define qualitative variable and quantitative variable, give any one example of each type.
- Describe briefly the main steps in the preparation of Frequency table from raw data.
- What is a cumulative frequency curve? How does it differ from the ordinary curve?
- Show that $\sum (x_i - a)^2 = \sum (x_i - \bar{x})^2 + n(\bar{x} - a)^2$ where 'a' is any arbitrary number.
- Differentiate between simple Arithmetic Mean and weighted Arithmetic Mean.
- What is meant by skewness? What are the measures of skewness, name them and also give formulae.
- The following values have been obtained from a frequency distribution of weights (lb) having 125 observation after making the substitution. $x = 16 + 5 U$, $\sum fu = -46$, $\sum fu^2 = 306$, $\sum fu^3 = -242$, and $\sum fu^4 = 1962$ check that either the distribution is Meso Kurtic or not.
- Compare the simple Index and composite Index.
- Show that Fisher's ideal Index satisfies the time reversal test.
- Show that coefficient of determination r^2 is $r^2 = \frac{a\sum y + b\sum x - (\sum y)^2 / n}{\sum y^2 - (\sum y)^2 / n}$
- Describe the properties of regression line.
- What is analysis of time series.
- Write brief note on semi average method.
- Define standard deviation of regression.

Solve the following questions.

Q:2 (a) Find (i) Arithmetic Mean and (ii) Geometric Mean of the series. 1, 3, 9, 27, 81.....3"

(b) The mean and Variance of variable "x" are 60 and 64 respectively. Find Mean and Variance of new variable if

(i) All the Values of x are increased by 20 points.

(ii) All the values of x are increased by 10%.

(5+5=10)

Q:3 (a) Obtain (i) Simple aggregative value index. (ii) An index of average value, for each year from the following data.

Years	Declared values	Values on the basis of 2010 values
2010	900	900
2011	1050	988
2012	1500	1000
2013	1700	832

(b) For n pairs of value of two variables when each variable is ranked in order (1 to n) show that

$$\text{coefficient of rank correlation is } r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}.$$

(5+5=10)

Q:4 (a) Compute Seasonal Indices by link Relative Method for the following data.

Year	Jan-March	April-June	July-Sep	Oct-Dec.
2001	80	101	126	92
2002	85	110	123	90
2003	83	99	125	94
2004	86	102	129	92

(b) Fit a Straight line $y = a + bx$ from the following results for the year 1948-58 (Both inclusive)

$$\sum x = 0, \sum y = 438.9, \sum x^2 = 110$$

$$\sum xy = -84.4$$

(6+4=10)

Find out the estimated trend values for 1960 and 1961.



Q.1. Answer the following short questions. (15x2=30)

- (i) Define Descriptive and Inferential statistics
- (ii) Define a variable. Differentiate between discrete and a continuous variable.
- (iii) What is meant by frequency distribution and cumulative frequency distribution.
- (iv) Define the statistical term histogram and frequency polygon.
- (v) What is a measure of central tendency? What are its desirable qualities?
- (vi) Show that $\sum(x_i - \bar{x}) = 0$
- (vii) What is coefficient of variation? What purpose does it serve?
- (viii) The first four moments of a distribution about $x=2$ are 1, 2.5, 5.5 and 16 calculate the first four moments about mean and calculate β_1 and β_2
- (ix) Compare the simple aggregative and weighted aggregative index numbers.
- (x) Prove that fisher's ideal index satisfies the factor reversal test.
- (xi) Show that standard deviation of regression $s_{y.x} = \sqrt{\frac{\sum y^2 - \frac{(\sum y)^2}{n}}{n-2}}$
- (xii) Define rank correlation, which quantity is added to $\sum d^2$ for tied ranks.
- (xiii) What is meant by coefficient of determination?
- (xiv) Distinguish between the secular and seasonal variation in time series analysis.
- (xv) Define detrending and deseasonalization.

Solve the following questions.

Q:2 (a) The following data relate to size of shoes sold at a store during a given week. Find the median and the modal size of shoes also find D_9 and P_{70} . (6+4=10)

Size of Shoes	5	$5\frac{1}{2}$	6	$6\frac{1}{2}$	7	$7\frac{1}{2}$	8	$8\frac{1}{2}$	9	$9\frac{1}{2}$
No of Pairs	3	7	20	30	60	50	25	17	8	2

(b) Show that $\text{var}(x+a) = \text{var}(x)$ where 'a' is some constant.

Q:3 (a) Show that correlation coefficient is independent of change of origin and scale i.e. $r_{xy} = r_{uv}$ if

$$u = \frac{x-a}{h}, v = \frac{y-b}{k} \quad (4+6=10)$$

(b) Construct fisher's ideal index number for 2011, taking 2010 as base

Items	Quantity		Value	
	2010	2011	2010	2011
A	110	175	600	1000
B	90	120	400	800
C	70	85	200	350

Q:4 (a) Compute the seasonal indices by ratio to trend method using least square line $y=119.56+2.76x$ by taking origin in the centre and unit of x being half quarter.

(b) Deseasonalize the values of years 1999 and 2000 (7+3=10)

QUARTERS				
Year	I	II	III	IV
1997	72	98	79	106
1998	79	122	101	143
1999	94	141	128	160
2000	125	143	135	187

Q.3 (a) Calculate the coefficient of correlation from the following data.

X	3	4	5	6	7	8	9	10	11	12
Y	25	24	20	20	19	17	16	13	10	6

(4+6=10)

Q.3 (b) Construct index numbers from the following data by applying (i) Laspeyre's method (ii) Paasche's method.

Commodities	Base Year		Current Year	
	Prices	Quantities	Prices	Quantities
A	6	50	10	56
B	2	160	2	120
C	4	60	6	60
D	10	30	12	24
E	8	40	12	36

Q.4 (a) Compute Seasonal Indices by link-relative method from the following data.

(6+4=10)

Year	Quarters			
	I	II	III	IV
2001	112	125	129	110
2002	119	132	147	115
2003	120	142	150	118
2004	128	151	162	125

(b) Fit a Straight line $y = a + b.x$ from the following results

$$\sum x = 0, \sum y = 438.9, \sum x^2 = 110$$

$$\sum xy = -84.4$$

for the year 1948-58 (Both inclusive) Find out the estimated trend values for 1960 and 1961.

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED****Q.1. Give short answer to the following concepts: (6x5=30)**

- i) Prove the property
 a) $Var(c) = 0$
 b) $Var(X - c) = Var(X)$, where c is a constant.
- ii) Differentiate between following terms by giving at least two examples.
 a) Descriptive and Inferential Statistics
 b) Ratio and Interval Scale
- iii) What is meant by a Histogram? Draw a histogram for the distribution of earnings (Rs.) given below

Earnings	180 - 184	185 - 189	190 - 194	195 - 199	200 - 204	205 - 209	210 - 214
Workers	10	24	36	40	29	23	8

Following table gives the production of cotton cloth (in million meters) for Pakistan from 1954 to 1962. Find index numbers by taking average of 1958, 1959 and 1960 as base.

Year	1954	1955	1956	1957	1958	1959	1960	1961	1962
Prod.	282	389	438	470	511	555	564	630	662

The number of cars crossing a certain bridge in a big city in 10 intervals of five minutes each was recorded as follows

25, 15, 18, 30, 20, 20, 12, 9, 16, 15, 23, 8

Calculate the arithmetic mean, median, geometric mean, mode and the harmonic mean.

The ages of 30 patients admitted to a certain hospital during a particular week are listed below. Construct a stem-and-leaf display.

48, 31, 54, 37, 18, 64, 61, 43, 40, 71, 51, 25, 65, 53, 42, 39, 62, 74, 48, 29, 67, 30, 49, 68, 15, 57, 26, 27, 58.

Answer the following questions.

Q2 Calculate the first four moments about the mean for the following data. (3x10=30)

x	1	2	3	4	5	6	7	8	9
f	1	6	8	15	20	12	9	5	2

Q3. Given the following, construct Marshall- Edgeworth and Fisher's Ideal index number for 1964, taking 1957 as base year.

Commodity	1957		1964	
	Price	Quantity	Price	Quantity
Rice	9.3	100	4.5	90
Wheat	6.4	11	3.7	19
Jawar	5.1	5	2.7	3

Q4. Plot the following data as a time series. Compute 4-quarter centered moving average trend and show it on the graph.

Year	Quarters			
	1	2	3	4
2000	102	71	47	98
2001	125	106	73	231
2002	281	229	209	488
2003	484	447	457	966